



St. Anthony's College
INFORMATION TECHNOLOGY DEPARTMENT
San Jose, Antique

**AICS Application Pre-Screening with Automated Document Verification and
Distribution Notification System for the Provincial Social Welfare and
Development Office (PSWDO)**

A Capstone Project

Presented to

The Faculty of the Information Technology Department

St. Anthony's College

In Partial Fulfillment

Of the Requirements for the Degree

Bachelor of Science in Information Technology

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CHAPTER III

PROJECT DESCRIPTION AND METHODOLOGY

This chapter gives the general description of developing "AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO)", including the methodology, participants in the research, the environment of the study, and the design of the system. It outlines the tools used to collect data for the study and the phases used in developing the system.

3.1. Description

3.1.1. System Overview

The capstone project, "AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO)," focuses on improving the processing and management of Assistance to Individuals in Crisis Situation (AICS) applications in the Province of Antique. The system is intended for residents of Antique who apply for AICS assistance and for authorized PSWDO personnel responsible for processing and managing applications. It aims to address problems related to manual document verification, applicant information encoding, duplicate records, application tracking, and the identification of cases that may require earlier attention.



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The system provides a web-based platform that supports online application submission, documentary requirement uploads, application tracking, SMS notifications, and an AI-assisted chatbot for applicants. For PSWDO personnel, the system supports application review, automated pre-screening, AI-enhanced document verification, duplicate detection, smart prioritization, distribution management, geo-mapping, analytics, dashboards, reports, audit trails, and centralized records management.

Participants of the system include AICS applicants and authorized PSWDO personnel. Applicants benefit from a more accessible and organized process for submitting requirements and monitoring application status while PSWDO personnel use the system to review, verify, prioritize, and manage AICS applications and assistance distributions. The Municipal Social Welfare and Development Offices (MSWDOs) are not actual system users and only receive distribution schedules and lists of approved beneficiaries from the PSWDO through email for information dissemination.

The capstone project development began in June 2026 with the initiation of the midyear academic class for Academic Year 2026-2027. The team plans to complete the project by December 2026, which marks the end of the first academic semester. During this period, the project covers the design, development, testing, implementation, and evaluation of the proposed system.



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The study will be conducted and implemented at the Provincial Social Welfare and Development Office (PSWDO) in the Province of Antique. The implementation is limited to AICS services within the province. Final assessment, validation, approval, rejection, and decisions regarding the type or amount of assistance remain under the authority of authorized PSWDO personnel.

Agile Scrum methodology is the chosen process for developing the system due to its iterative development, collaboration, and flexibility through short cycles known as sprints. The proposed AICS system is conceptualized using the Input-Process-Output (IPO) structure with an evaluation stage, as illustrated in Figure 1 below.

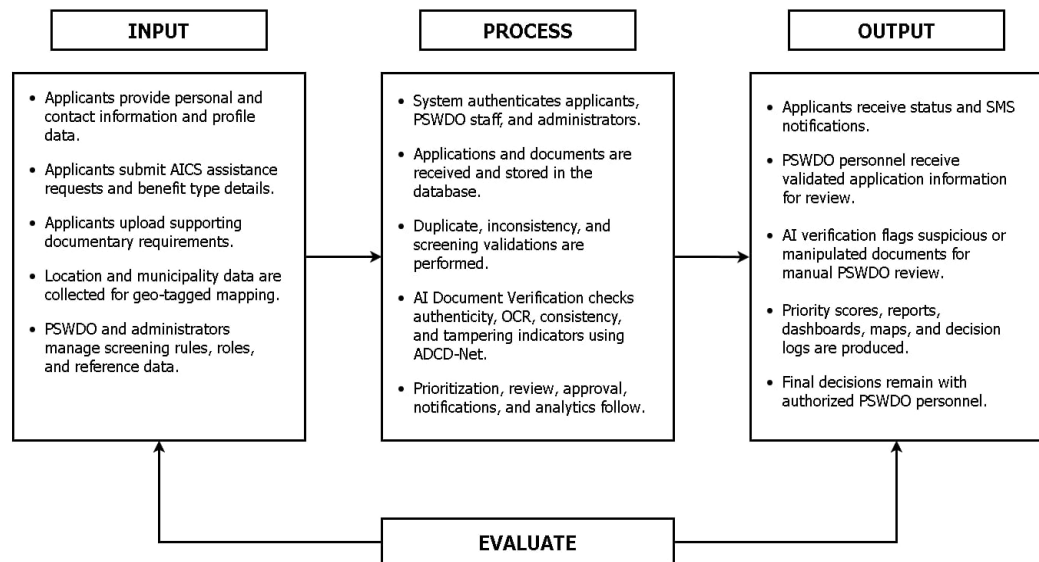


Figure 1. Conceptual Framework of AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO)



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The Input-Process-Output (IPO) model represents the overall flow of the proposed system. At the input stage, applicants provide personal and contact information, assistance request details, documentary requirements, and municipality and barangay information. Authorized PSWDO personnel and administrator provide user credentials, screening rules, and reference data. During the process stage, the system authenticates users, stores application data in the centralized database, and performs automated pre-screening, duplicate and inconsistency checking, AI-enhanced document verification, and smart prioritization. Documents with incomplete, inconsistent, unreadable, altered, or potentially tampered information are flagged for further review by PSWDO personnel.

At the output stage, applicants can view application status and receive SMS notifications regarding missing requirements, corrections, decisions, and assistance distribution schedules. PSWDO personnel can access verification results, priority information, dashboards, reports, geo-maps, analytics, audit trails, and distribution records. These outputs are evaluated during the evaluation stage to consolidate feedback from both applicants and PSWDO personnel, further refine the system's features and functionalities, and enhance its overall performance in supporting AICS application processing and assistance distribution.



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3.1.2. System Features

The AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO) offers integrated features that work together to improve the processing, verification, monitoring, and management of AICS applications in the Province of Antique.

Online Application Submission - Allows applicants to create an account, complete their profile, submit an AICS application, and upload the required documentary requirements through the web-based system. Application information and uploaded documents are stored in a centralized database for easier access and management.

Automated Pre-Screening - Checks whether the required applicant information and documentary requirements have been submitted. The system identifies incomplete or missing information before the application proceeds to further review.

AI-Enhanced Document Verification – Assists in checking uploaded documentary requirements for incomplete, inconsistent, unreadable, altered, or potentially tampered information. Documents that require further verification are flagged for review by authorized PSWDO personnel.



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Duplicate Detection - Compares applicant and beneficiary information with previous AICS records to identify possible duplicate applications or matching records that require further verification.

Smart Prioritization - Classifies AICS applications according to beneficiary information and identified priority categories. This feature assists PSWDO personnel in identifying cases that may require earlier attention.

Application Tracking - Allows applicants to monitor the current status of submitted AICS applications. PSWDO personnel can also monitor and update application records as the application moves through the review process.

SMS Notification - Sends applicants updates regarding application status, missing requirements, required corrections, application decisions, and assistance distribution schedules.

AI-Assisted Chatbot - Provides applicants with guidance regarding AICS application procedures, documentary requirements, application status descriptions, and system navigation. The chatbot serves only as an assistance tool and does not determine eligibility or approve applications.

Distribution Management – Allows authorized PSWDO personnel to manage assistance schedules, claiming requirements, distribution details, and records indicating whether approved assistance has already been released or claimed.



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Geo-Mapping - Displays AICS application information according to municipalities or barangays. This feature helps PSWDO personnel identify areas with higher numbers of applications and assistance needs.

Analytics Dashboard and Reports - Provides summarized information about AICS applications, beneficiaries, priority levels, assistance types, application statuses, and assistance distribution records. The dashboard supports monitoring and reviewing AICS operations.

Audit Trail - Records important system activities such as application submissions, document uploads, record modifications, status updates, approvals, notifications, and assistance distribution activities.

Centralized Records Management - Provides organized storage and retrieval of applicant profiles, applications, documentary requirements, application statuses, assessment records, and assistance distribution records.

3.1.3. System Goals and Objectives

The primary goal of the system is to address the challenges in processing and managing AICS applications, particularly manual document verification, applicant information encoding, duplicate records, application monitoring, and identification of priority cases, which are concerns of the PSWDO in the Province of Antique. To resolve these problems, the researchers developed the AICS Application Pre-Screening with Automated Document Verification and Distribution



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Notification System for the Provincial Social Welfare and Development Office (PSWDO), a web-based platform designed to improve application processing, document verification, monitoring, and assistance distribution.

The specific goals of the system are to:

- Create a centralized database for managing applicant profiles, AICS applications, documentary requirements, application statuses, and assistance distribution records.
- Design a web-based system that allows applicants to submit and track applications, receive SMS notifications, and use an AI-assisted chatbot, and for PSWDO personnel to review applications, manage distributions, and access dashboards and geo-mapping.
- Develop automated pre-screening, AI-enhanced document verification, duplicate detection, and smart prioritization based on beneficiary information and uploaded documentary requirements.
- Implement and evaluate the system in the PSWDO based on the ISO/IEC 25010 software quality standards in terms of functionality, efficiency, usability, reliability, and security.

3.1.4. System Architecture

The architecture of the AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO) shown in Figure 2 below, web-



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based platform, centralized database, AI document verification, email, and SMS notification services. The system consists of three main user components: Applicant, PSWDO Personnel, and Administrator.

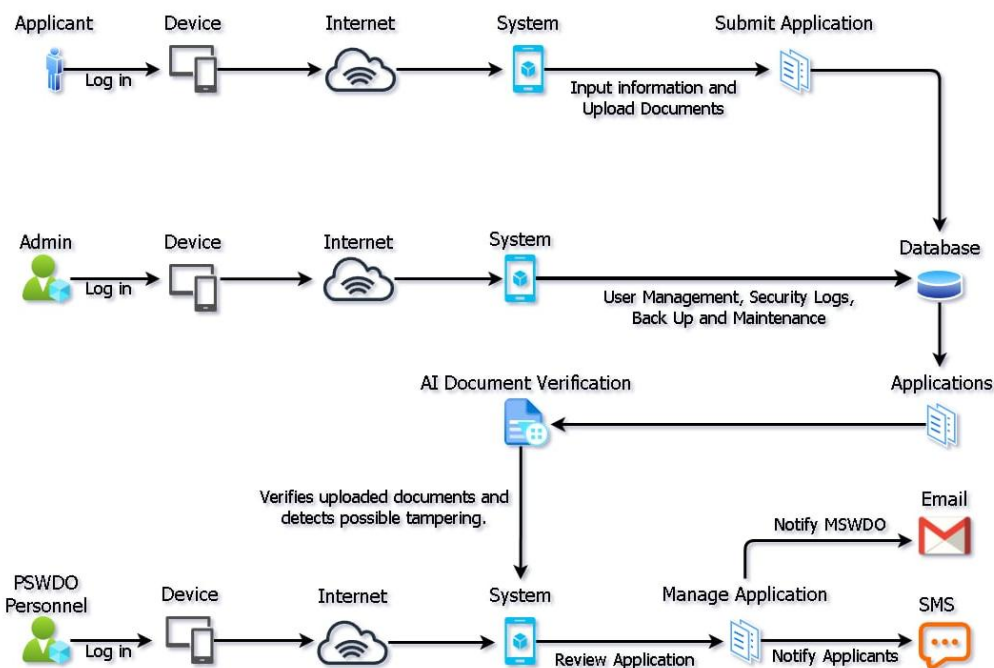


Figure 2. AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO) System Architecture

The first layer is the Applicant layer, which allows residents of Antique to access the system using a device with an internet connection. Applicants log in to the system, provide the required information, upload documentary requirements, and submit an AICS application. The submitted application and documents are stored in the centralized database for processing and review.



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Secondly, the PSWDO Personnel layer allows authorized personnel to access the system through secure login credentials. Submitted applications are processed through the AI Document Verification component, which assists in checking uploaded documents and detecting possible inconsistencies or tampering. PSWDO personnel review and manage applications based on the information and verification results provided by the system. SMS notifications are sent to applicants regarding application updates, while email notifications are sent to the Municipal Social Welfare and Development Offices (MSWDOs) for information dissemination.

Lastly, the Administrator layer allows authorized administrators to access and maintain the system. Administrators are responsible for user management, security logs, system backup, and maintenance. The centralized database stores application records, user information, and other system data required by the Applicant, PSWDO Personnel, and Administrator components. Final assessment, validation, approval, rejection, and decisions regarding the type or amount of assistance remain under the authority of authorized PSWDO personnel.

3.2. Methods

3.2.1. Development Methodology

Agile Scrum is an iterative development method that enables teams to adapt to changing requirements while improving the system throughout development. The development process is divided into periods called sprints, that



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commonly last from 2 to 4 weeks. Each sprint focuses on completing specific system functions that can be tested, reviewed, and improved based on feedback. This approach allows the development team to divide a large project into manageable tasks, identify problems earlier, and make necessary changes before the entire system is completed.

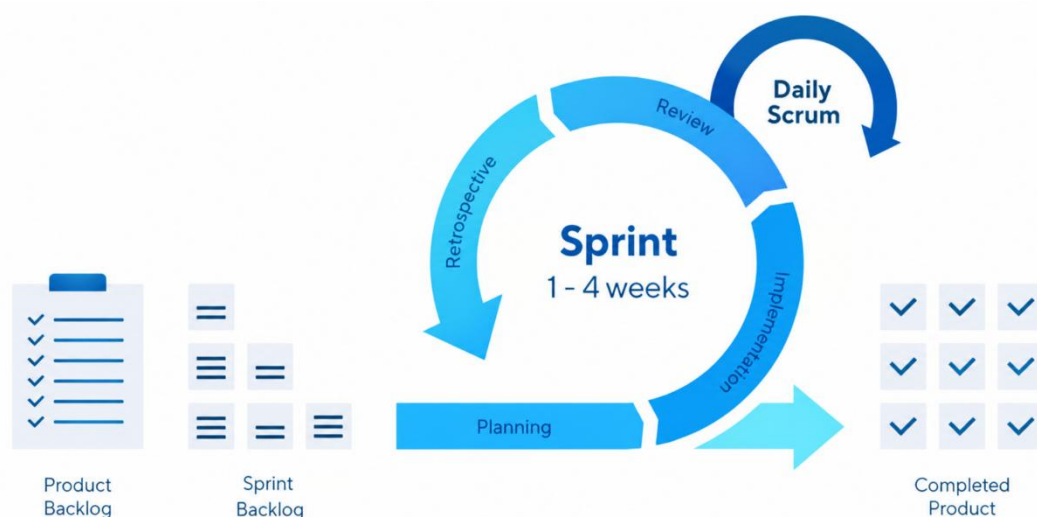


Figure 3. Agile Scrum Methodology

The researchers adapted the Agile Scrum methodology for the development of the "AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO)" because of the different functions that need to be developed and integrated into one web-based system. The development process uses Agile Scrum to progressively build and test essential functions, including online application submission, automated pre-screening, AI-enhanced



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document verification, duplicate detection, smart prioritization, application tracking, SMS notifications, distribution management, geo-mapping, dashboards, reports, and centralized records management.

The iterative approach allows each feature to be developed, tested, reviewed, and improved before complete system integration. Collaboration among the developers, designers, and PSWDO personnel also helps ensure that the system functions are aligned with the actual AICS application process and operational requirements of the office. Feedback gathered during each development cycle can be used to improve system functionality, usability, reliability, and overall performance. Through Agile Scrum, development risks can be reduced while allowing necessary adjustments throughout the design, development, testing, implementation, and evaluation of the proposed system.

3.2.2. System Development Life Cycle Phases

- **Product Backlog** - Identified the major problems in the existing AICS application process, including manual document verification, applicant information encoding, duplicate records, difficulty in tracking application status, and identifying priority cases. The foundation of the system was established by defining the needed functions for a centralized web-based platform that supports AICS application processing and assistance distribution.



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- **Sprint Backlog** - This phase is divided into iterative cycles called "Sprints," where each sprint is classified into the following sub-phases:
 - **Planning** - The system scope, requirements, and major features were identified, including online application submission, automated pre-screening, AI-enhanced document verification, duplicate detection, smart prioritization, application tracking, SMS notifications, distribution management, geo-mapping, dashboards, reports, audit trails, and centralized records management.
 - **Implementation and Daily Scrum** - The actual development of the system's main functions was conducted during this phase. The development team worked on the applicant interface, PSWDO personnel functions, administrator functions, database integration, automated pre-screening, document verification, application tracking, notifications, and other system features. Daily Scrum meetings were conducted to monitor progress, identify development issues, and coordinate tasks among the developers and designers.
 - **Review** - Completed system features were presented and reviewed at the end of each sprint. Feedback from PSWDO personnel was gathered to determine whether the developed functions were aligned with the existing AICS process and operational requirements of the office.



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- **Retrospective** - The development team evaluated the work completed during each sprint to identify problems encountered during development. Necessary improvements were applied to the system features, interface, processing functions, and overall performance before proceeding to the next development cycle.
- **Deployment** - The system deployment process will be conducted at the Provincial Social Welfare and Development Office (PSWDO) in the Province of Antique. The system will be made accessible to authorized PSWDO personnel and AICS applicants for implementation and evaluation. The deployed system will be evaluated based on the ISO/IEC 25010 software quality standards in terms of functionality, efficiency, usability, reliability, and security.

3.2.3. Data Gathering Instrument

The main goal of data collection is to provide additional insights, along with new data and information needed to develop the AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO). The developers used publicly available sources to review existing research on social welfare services, document verification, and government assistance systems. The study was conducted at the Provincial Social Welfare and Development Office (PSWDO) in the Province of Antique and residents who apply for AICS assistance.



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To gather data, the developers conducted surveys and an open-ended questionnaire. The developers created a paper-based survey questionnaire composed of checkboxes, which was distributed to 31 AICS applicants and 11 PSWDO personnel. The survey questionnaire included questions that could be answered with Yes or No, as well as questions related to the existing AICS application process, document submission, application monitoring, duplicate records, and the proposed system. The researchers aimed to assess the participants' experiences with the current AICS process, identify existing difficulties encountered by applicants and PSWDO personnel, and determine the system features needed to improve application processing and management.

Table 1. AICS Applicants and PSWDO Personnel Survey Results

Applicants	Response		PSWDO Personnel	Response	
	Yes	No		Yes	No
Q1	31	0	Q1	11	0
Q2	18	13	Q2	8	3
Q3	18	13	Q3	11	0
Q4	25	6	Q4	11	0
Q5	22	9	Q5	8	3
Q6	27	4	Q6	9	2
Q7	28	3	Q7	6	5
Q8	30	1	Q8	11	0
Q9	30	1	Q9	10	1
Q10	30	1	Q10	11	0
Total	259	51	Total	96	14
Total	83.55%	16.45%	Total	87%	12.73%

The table shows survey results from 31 AICS applicants and 11 PSWDO personnel. The questionnaire consists of 10 questions, each with a "Yes" or "No"



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answer. With a total of 310 responses, the AICS applicants recorded 259 "Yes" responses and 51 "No" responses, which resulted in 83.55% Yes and 16.45% No. The PSWDO personnel recorded 96 "Yes" responses and 14 "No" responses out of 110 responses, resulting in 87.27% Yes and 12.73% No.

Grand Overall Survey Results

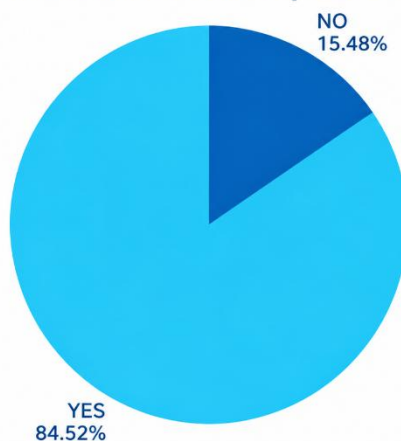


Figure 4. Survey Results in a Pie Chart

The pie chart in Figure 4 shows the overall distribution of answers from both AICS applicants and PSWDO personnel. The results show that 84.52% responded "Yes" and 15.48% responded "No."

3.2.4. Validation Process

To ensure the system performs correctly and efficiently and meets user experience expectations before full deployment, the development has undergone validation.



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3.3. Key Deliverables and Performance Matrix

3.3.1. Deliverables

The deliverables for the

3.3.2. Performance Matrix

The Performance Matrix for the

3.4. Project Timeline

3.4.1. Project Schedule Overview

The project schedule for the AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO) covers June 2026 to December 2026, with a duration of a seven-month development period. The Planning Phase serves as the initial step of project development, which begins in June when the team identifies the project objectives, scope, system requirements, and resources needed to build the project foundation. The Design Phase, conducted in July, requires the team to develop the system architecture, interface layouts, wireframes, database structure, and process flows that serve as the blueprint for the web-based system. The Development Phase, which follows the design stage from August to September, serves as the main development period when the developers create and integrate the system's major functions.



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The project involves the development of essential functionalities, including online application submission, automated pre-screening, AI-enhanced document verification, duplicate detection, smart prioritization, application tracking, SMS notifications, distribution management, geo-mapping, dashboards, reports, audit trails, and centralized records management. In October 2026, the Testing Phase takes place to assess the system's functionality, document verification, database operations, user interface, notifications, and overall performance. The Deployment and Maintenance Phase, from November to December 2026, marks the implementation of the system at the Provincial Social Welfare and Development Office (PSWDO) in the Province of Antique, where the system is prepared for actual use and evaluation. Maintenance activities are also conducted to address identified issues, apply necessary improvements, and ensure that the system operates properly and meets its intended purpose.

3.4.2. Gantt Chart

A Gantt chart is a visual project management tool that illustrates a project's schedule by showing tasks over a specific period of time. It enables the researchers to monitor the start and end of each activity, task duration, project progress, and upcoming deadlines. It displays the entire development process of the AICS Application Pre-Screening with Automated Document Verification and Distribution Notification System for the Provincial Social Welfare and Development Office (PSWDO), which all development phases are organized and



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monitored. The chart shows the project timeline from June 2026 to December 2026, covering a seven-month development period, as shown in Figure 5.

The researchers conducted the Planning Phase during June 2026, where the project objectives, scope, system requirements, needed resources, and task assignments were identified. The Design Phase was conducted in July 2026 and focused on the development of the system architecture, database structure, user interface layouts, wireframes, and process flows. The Development Phase took place from August to September 2026, during which the team developed and integrated the main system features, including online application submission, automated pre-screening, AI-enhanced document verification, duplicate detection, smart prioritization, application tracking, SMS notifications, distribution management, geo-mapping, dashboards, reports, audit trails, and centralized records management.

	JUNE				JULY				AUGUST				SEPTEMBER				OCTOBER				NOVEMBER				DEC
	Weeks 1 - 4				Weeks 5 - 8				Weeks 9 - 12				Weeks 13 - 16				Weeks 17 - 20				Weeks 21 - 24				Week 25
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Planning Phase																									
Design Phase																									
Development Phase																									
Testing Phase																									
Deployment & Maintenance																									

Figure 5. AICS Gantt Chart



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The Testing Phase was conducted in October 2026 to identify errors and assess the functionality, performance, database operations, document verification, notifications, and user interface of the system. Necessary improvements and debugging activities were also performed during this period. The final phase, Deployment and Maintenance, was scheduled from November to December 2026. During this phase, the system is implemented at the Provincial Social Welfare and Development Office (PSWDO) in the Province of Antique, prepared for actual use and evaluation, and maintained to address identified issues and further improve system performance.



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